

CSA0503-DATABASE MANAGEMENT SYSTEMS

EXPERIMENT 14 - TCL COMMANDS – COMMIT, SAVEPOINT, ROLLBACK

1)

```
mysql> CREATE TABLE CLASS
-> (
->     name VARCHAR(20),
->     id INT
-> );
Query OK, 0 rows affected (0.06 sec)

mysql> INSERT INTO CLASS VALUES
-> ('alpha', 1),
-> ('beta', 2),
-> ('charlie', 3),
-> ('delta', 4),
-> ('echo', 5);
Query OK, 5 rows affected (0.04 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM CLASS;
+-----+-----+
| name   | id   |
+-----+-----+
| alpha  | 1    |
| beta   | 2    |
| charlie| 3    |
| delta  | 4    |
| echo   | 5    |
+-----+-----+
5 rows in set (0.02 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE CLASS
-> SET name = 'bravo'
-> WHERE id = 5;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> SELECT * FROM CLASS;
+-----+-----+
| name  | id  |
+-----+-----+
| alpha | 1   |
| beta  | 2   |
| charlie | 3   |
| delta | 4   |
| bravo | 5   |
+-----+-----+
5 rows in set (0.00 sec)

mysql> SAVEPOINT A;
Query OK, 0 rows affected (0.00 sec)

mysql> INSERT INTO CLASS VALUES('uppal', 6);
Query OK, 1 row affected (0.00 sec)

mysql> SELECT * FROM CLASS;
+-----+-----+
| name  | id  |
+-----+-----+
| alpha | 1   |
| beta  | 2   |
| charlie | 3   |
| delta | 4   |
| bravo | 5   |
| uppal | 6   |
+-----+-----+
6 rows in set (0.00 sec)

mysql> SAVEPOINT B;
Query OK, 0 rows affected (0.00 sec)

mysql> INSERT INTO CLASS VALUES('balu', 7);
Query OK, 1 row affected (0.00 sec)
```

```
mysql> SELECT * FROM CLASS;
+-----+-----+
| name  | id  |
+-----+-----+
| alpha | 1   |
| beta  | 2   |
| charlie | 3   |
| delta | 4   |
| bravo | 5   |
| uppal | 6   |
| balu  | 7   |
+-----+-----+
7 rows in set (0.00 sec)

mysql> SAVEPOINT C;
Query OK, 0 rows affected (0.00 sec)

mysql> ROLLBACK TO SAVEPOINT B;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT * FROM CLASS;
+-----+-----+
| name  | id  |
+-----+-----+
| alpha | 1   |
| beta  | 2   |
| charlie | 3   |
| delta | 4   |
| bravo | 5   |
| uppal | 6   |
+-----+-----+
6 rows in set (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)
```

RESULT:

The commands SQL COMMIT, ROLLBACK and SAVEPOINT are implemented and result is verified